Digital Logic Design (DLD)

Mid Question Answer

Lecture – 1

1. What is Digital Logic Design?

Ans: **The** digital logic design **is a system in electrical and computer engineering that** uses simple numerical values to produce input and output operations. **Assist in** developing cell phones, computers, and related personal electronic devices**.**

2. How type of Numerical Representations?

Two ways of representing the numerical value of quantities: i. Analog and
ii. Digital

i. **Analog Representations:**
In analog representation a quantity is represented by a **continuously variable, proportional indicator.**

Analog = Continuous

ii. Digital Representations:
In digital representation the quantities are represented not by **continuously variable indicators but by symbols called *digits.*** As an example, consider the **digital clock.**
Digital = Discrete (step by step)

3. Example 1: Which of the following involve analog quantities and which involve digital quantities?

(a) Ten-position switch
(b) Current flowing from an electrical switch
(c) Temperature of a room
(d) Sand grains on the beach
(e) Automobile fuel gauge

Solution:
(a) Digital
(b) Analog
(c) Analog
(d) Digital
(e) Analog, if needle type; digital, if numerical readout or bar graph display

4. Define Digital and Analog System.

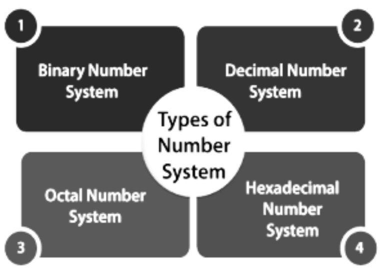
<u>**Digital System:**</u> A **digital system** is a **combination of devices** designed to **manipulate logical information or physical quantities** that are represented in **digital form**

<u>**Analog System:**</u> An **analog system** contains devices that **manipulate physical quantities** that are represented in **analog form**.

5. Advantages of Digital Techniques.

1. Digital systems are generally easier to design.
2. Information storage is easy.
3. Accuracy and precision are easier to maintain throughout the system.
4. Operation can be programmed.
5. Digital circuits are less affected by noise.
6. More digital circuitry can be fabricated on IC chips.

6. Type of Number System. Page (7-10)



i. Binary Number: Base 2. Carries only two digits, like that 0 or 1.

ii. Decimal Number: Base 10. Contain Ten digits, 0 to 9.

iii. Octal Number: Base 8. Contain Eight digit, 0 to 7.

iv. Hexadecimal Number: Base 16. Contain total 16 symbols like 0 to 15 (0,1,2,3,4,5,6,7,8,9, A, B, C, D, F)

7. Number Base Conversion.

Ans: Page 11-19

Lecture – 2

8. What is Binary Codes?

Ans: In the coding, when **numbers, letters or words are represented** by a **specific group of symbols**, it is said that the number, letter or word is being **encoded**.

9. Advantages of Binary code.

1. Binary codes are **suitable for the computer applications.**
2. suitable for the digital communications.
3. the **analysis and designing of digital circuits much easier.**
4. Since only 0 & 1 are being used, **implementation becomes easy.**

10. Classification of Binary Codes

- Binary Coded Decimal (BCD) Code
- Gray Code
- Excess-3 Code
- Alphanumeric Codes
- Error Detecting Codes
- Error Correcting Codes

11. Explain and Conversion BCD Code.

In this code each **decimal digit is represented by a 4-bit binary number.** BCD is a way to express each of the decimal digits with a binary code. BCD code **only first ten** (0000 to 1001) of these are **used.**

Decimal	0	1	2	3	4	5	6	7	8	9
BCD	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

12. Advantages and Disadvantages of BCD Code.

Advantage:

- It is very **similar to decimal** system.
- We **need to remember binary equivalent** of decimal numbers 0 to 9 only.

Disadvantages:

- The **addition and subtraction** of BCD have different rules
- The BCD **arithmetic** is little more complicated.
- BCD **needs a greater number of bits** than binary to **represent the decimal** number.

13. Define Gray Code.

Ans: The **Gray Code** is a sequence of binary number systems, which is also known as **Reflected Binary Code (RBC)**.

Conversion -> i. Binary to Gray code and ii. Gray to Binary code.

Ans: i. page (3-4) ii. Page (4-5)

14. Explain Gray code characteristics.

- In Gray code, **two consecutive values are differed by one bit** of binary digits
- Binary number is **converted to Gray code to reduce the switching operation.**

15. Conversion Binary to Exces-3 code.

Ans: page (7-8)

- **Methods for Finding Excess-3 code of Binary Number.**

Method 01:

- We **find the decimal number** of the given binary number.
- Then we **add 3 in each digit** of the decimal number.
- Now, we **find the binary code of each digit** of the newly generated decimal number.

Method 02:

- We can also **add 0011 in each 4-bit BCD code** of the decimal number for getting excess-3 code.

16. Self-Complementary Property.

Ans: Page (09)

• A self-complementary binary code is a code which is **always complimented in itself.** By **replacing the bit 0 to 1 and 1 to 0** of a number, we find the **1's complement** of the number.

• If we perform the **1's complement of excess-3 of a decimal number**, it will be **equal** to the **excess-3 code of the 9's complement of that decimal number.**

17. Difference Between Unicode and ASCII

Unicode Coding Scheme	ASCII Coding Scheme
It uses variable bit encoding according to the requirement. For example, UTF- 8, UTF-16, UTF-32	It uses 7-bit encoding. As of now, the extended form uses 8-bit encoding.
It is a standard form	It is not a standard all over the world.
People use this scheme all over the world	It has only limited characters hence, it cannot be used all over the world.
It has more than 128,000 characters.	In contrast, it has only 256 characters

Lecture – 03

18. Define De Morgan's Theorems.

Ans: There are two basic theorems of great importance in Boolean Algebra, which are **De Morgan's First Law** and **De Morgan's Second Law.** These are also called De Morgan's Theorems.

i. De Morgan's First Law: (P.Q)' = (P)' + (Q)'

ii. De Morgan's Second Law: (P + Q)' = (P)'. (Q)'

19. Applications of Boolean Algebra

- Digital Logic Design
- Algorithm Design
- Telecommunications
- Artificial Intelligence (AI)
- Electrical Engineering

20. Math page (6-10)

The basic laws of Boolean Algebra are summarized in the table below:

Law	OR Form	AND Form
1. Identity Law	$P + 0 = P$	$P \cdot 1 = P$
2. Idempotent Law	$P + P = P$	$P \cdot P = P$
3. Commutative Law	$P + Q = Q + P$	$P \cdot Q = Q \cdot P$
4. Associative Law	$P + (Q + R) = (P + Q) + R$	$P \cdot (Q \cdot R) = (P \cdot Q) \cdot R$
5. Distributive Law	$P + (Q \cdot R) = (P + Q) \cdot (P + R)$	$P \cdot (Q + R) = (P \cdot Q) + (P \cdot R)$
6. Inversion Law	$(A')' = A$	$(A')' = A$

Law	OR Form	AND Form
7. De Morgan's Law	$(P + Q)' = P' \cdot Q'$	$(P \cdot Q)' = P' + Q'$
8. Complement Law	$P + P' = 1$	$P \cdot P' = 0$
9. Domination Law	$P + 1 = 1$	$P \cdot 0 = 0$
10. Double Negation Law	$(P')' = P$	$(P')' = P$
11. Absorption Law	$P + (P \cdot Q) = P$	$P \cdot (P + Q) = P$

Lecture – 04

21. Advantage and Disadvantages of K-Map.

Advantages:

a) Makes Logic Simpler
b) Minimizes Logic Gates
c) Reduce Errors
d) Time-Saving
e) No need for Boolean Laws

Disadvantages:

a) Limited to Fewer Variables
b) Not suitable for all functions
c) Space Limitations
d) Requires Careful Grouping

22. Math page (1-04).